

Name	: Mr. VIPAN KUMAR GARG	Age	: 68 Years
Lab No.	: 193146741	Gender	: Male
Ref By	: PGI CHANDIGARH	Reported	: 10/2/2025 1:12:33PM
Collected	: 7/2/2025 9:40:00AM	Report Status	: Final
A/c Status	: P	Processed at	: LPL-NATIONAL REFERENCE LAB
Collected at	: CHANDIGARH-IV	National Reference laboratory, Block E,	
	Booth Number-122,Sector 10, Inner Market	Sector 18, Rohini, New Delhi -110085	
	Panchkula		
	PANCHKULA		



Test Report

Test Name	Results	Units	Bio. Ref. Interval
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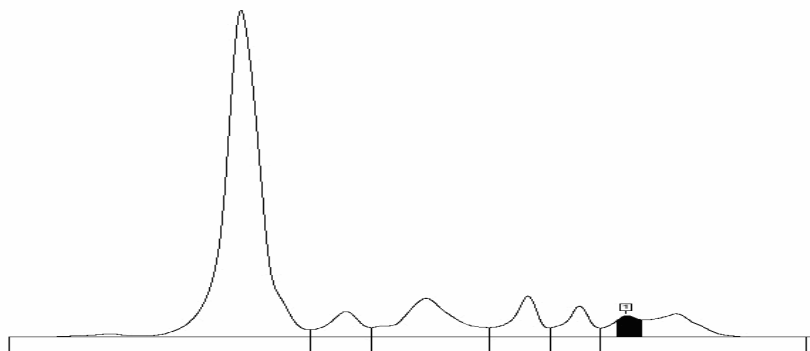
MULTIPLE MYELOMA MONITORING PANEL

PROTEIN ELECTROPHORESIS, SERUM

(Capillary Electrophoresis)

Protein, Total	7.20	g/dL	6.40 - 8.10
Albumin	4.69	g/dL	3.60 - 5.40
Alpha 1 globulin	0.32	g/dL	0.20 - 0.40
Alpha 2 globulin	0.80	g/dL	0.50 - 1.00
Beta 1 globulin	0.42	g/dL	0.50 - 1.10
Beta 2 globulin	0.30	g/dL	0.30 - 0.60
Gamma globulin	0.66	g/dL	0.70 - 1.50
A : G Ratio	1.87		0.90 - 2.00
M Spike	? 0.2	g/dL	

ID **193146741**



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Test Name	Results	Units	Bio. Ref. Interval
Interpretation			
Suspicious non-discrete protein band in gamma globulin region, possibly Monoclonal gammopathy. Advised: Immunofixation electrophoresis (IFE) to confirm the presence of monoclonal gammopathy and identify its immunological nature.			
Comments			
Serum Protein electrophoresis (SPE) is used to identify patients with Monoclonal gammopathies (Multiple Myeloma (MM), Waldenstrom's macroglobulinemia, AL Amyloidosis as well as premalignant conditions Monoclonal gammopathy of unknown significance(MGUS) and Smoldering Myeloma) and other serum protein abnormalities (Nephrotic syndrome, Alpha 1 antitrypsin disorder, and inflammatory processes associated with infection, liver diseases & autoimmune disorder). Monoclonal gammopathies indicate clonal expansion of plasma cells or mature B cells. Monoclonal gammopathies are present in 8% of geriatric patients and require further evaluation for monitoring & prognosis. SPE can be used for monitoring response to therapy, a decrease or increase of M spike by 0.5 g/dl is considered significant change.			
If SPE alone is used as initial diagnostic screen it will be able to detect approximately 88% of all MM, 66% of AL Amyloidosis and 56% of Light Chain Deposition Disease (LCDD) patients. SPE has good sensitivity in detecting intact monoclonal protein but has limited sensitivity in detecting monoclonal free light chains (FLC). Therefore, SPE alone will miss many patients that are only producing monoclonal FLC. These misses include many Light Chain MM, all Non Secretory MM and significant numbers of AL amyloidosis and LCDD patients. Combination of FLC assay with SPE & immunofixation improves the sensitivity to 99%. Thus, Multiple Myeloma screening panel which includes SPE, IFE and FLC should be done to screen patients with high clinical suspicion for MM.			

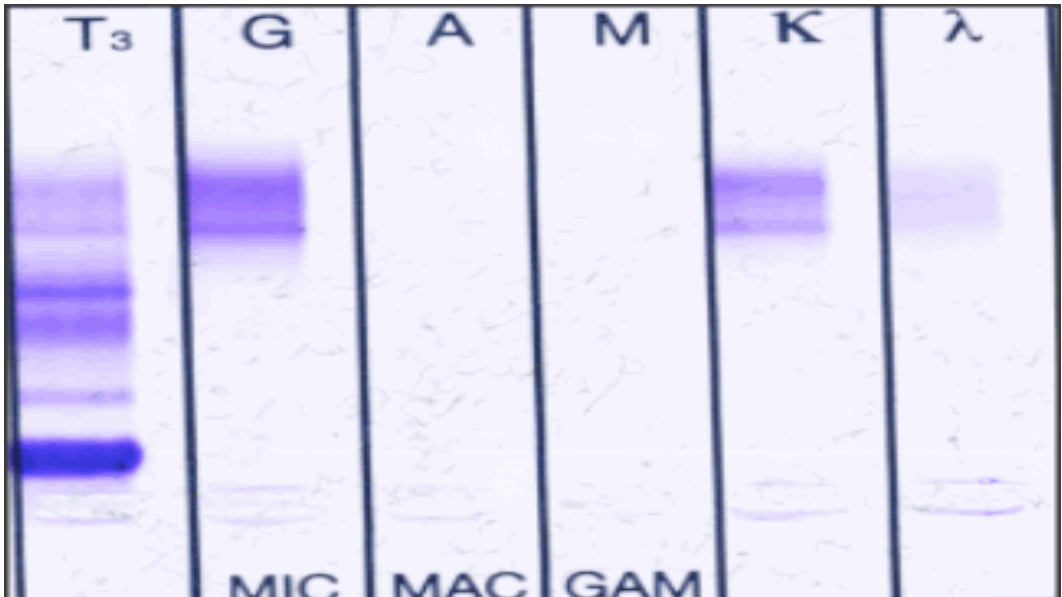
IMMUNOTYPING, SERUM



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
			

Interpretation:-Serum protein profile by high-resolution Agarose gel electrophoresis reveals the presence of a discrete protein band in gamma globulin region suggestive of 'M' spike

A 2nd protein band proximal to it is also seen in gamma globulin region could be isomer of the same.

Immunofixation electrophoresis (IFE) identifies the 'M' spike as IgG Kappa and its (?) isomer as IgG Kappa .

Note: Immunotyping is not a quantitative assay. If a monoclonal protein is identified, a serum protein electrophoresis assay is required for quantifying the abnormality.

Comment

The assay is used for immunotyping of monoclonal proteins which identifies the monoclonal immunoglobulin heavy-chain (gamma, alpha, mu) and/or light-chain type (kappa or lambda). It is generally recommended that both serum Protein electrophoresis (SPEP) and Immunotyping be used as a screening panel because it is more sensitive than SPEP. Immunotyping is not only recommended as part of the initial screening process but also for confirmation of complete response to therapy.

Usage

Identification of monoclonal immunoglobulin heavy and light chains
Documentation of complete response to therapy



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
BETA 2 MICROGLOBULIN, SERUM (CLIA)			
Beta-2 Microglobulin, Serum	7864	ng/mL	609.00 - 2366.00

- Note**
- False negative / positive results are observed in patients receiving mouse monoclonal antibodies for diagnosis or therapy.
 - Beta 2 Microglobulin values regardless of levels should not be interpreted as absolute evidence of the presence or absence of disease. All values should be correlated with clinical findings and results of other investigations.

Comment
As per International Staging System for Multiple Myeloma the level of Beta 2 microglobulin can be used for staging the disease:

STAGE	BETA 2 MICROGLOBULIN LEVEL IN (ng/mL)
Stage I	<3500
Stage II	3500-5500
Stage III	>5500

- Increased levels**
- * Lymphoproliferative disorders like Multiple myeloma, B cell Lymphoma and Chronic Lymphocytic Leukemia.
 - * Inflammatory disorders like Rheumatoid arthritis, SLE, Sjogren's syndrome and Crohn's disease.
 - * Renal Dysfunction

- Uses**
- * Prognostic indicator of Multiple myeloma and other hematopoietic malignancies
 - * An aid in the management of patients with Renal dysfunction and Rheumatoid arthritis
 - * Staging of Multiple Myeloma

IMMUNOGLOBULIN PROFILE, SERUM



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
Immunoglobulin IgG, Serum Immunoturbidimetry)	736.00	mg/dL	650.00 - 1600.00
Immunoglobulin IgM, Serum	<21.0	mg/dL	50.00 - 300.00
Immunoglobulin IgA, Serum	52.70	mg/dL	40.00 - 350.00

Comments:

Approximately 80% of serum immunoglobulin is IgG, 6% is IgM and 13% is IgA. High levels of IgM signify acute infections whereas IgG predominates in chronic infections. IgA is the predominant immunoglobulin in body secretions like saliva, sweat and colostrums.

Polyclonal increases are seen in:

- IgG: SLE, Chronic liver diseases, Infectious diseases and Cystic fibrosis
- IgM: Viral, bacterial and parasitic infections, Liver diseases, Rheumatoid arthritis, Scleroderma, Cystic fibrosis & heroin addiction
- IgA: Chronic liver diseases, Chronic infections, Autoimmune disorders, Sarcoidosis and Wiscott-Aldrich syndrome.

Monoclonal increases are seen in IgG & IgA Myelomas and IgM increases in cases of Waldenstroms macroglobulinemia

Decreased synthesis of IgG, IgM & IgA is seen in Congenital and Acquired Immunodeficiency diseases.

Decreased levels are seen in Protein losing enteropathies, Nephrotic syndrome and skin burns.



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
KAPPA / LAMBDA LIGHT CHAINS, FREE, SERUM (Immunoturbidimetry)			
Kappa, Free Light Chain	46.57	mg/L	3.30 - 19.40
Lambda, Free Light Chain	15.04	mg/L	5.71 - 26.30
Kappa/Lambda(FLC), Ratio	3.096		0.26 - 1.65

Interpretation

KAPPA	LAMBDA	KAPPA / LAMBDA RATIO	REMARKS
Normal	Normal	Normal	Normal sample. If serum electrophoretic pattern is normal, it is unlikely that patient has Monoclonal gammopathy.
Normal	Normal	Low/High	Suggestive of Monoclonal gammopathy with Bone Marrow suppression
Low	Low	Normal	Suggestive of Bone Marrow suppression
Low	Low	Low/High	Suggestive of Monoclonal gammopathy with Bone Marrow suppression
High	High	Normal	Suggestive of : • Renal Impairment • Overproduction of Polyclonal free light chain from inflammatory conditions • Biclinal gammopathy of different free light chain types
High	High	Low/High	Suggestive of Monoclonal gammopathy with Renal Impairment

Note: International Myeloma Working Group (IMWG-2014) has updated the criteria for the diagnosis of Multiple Myeloma (MM) which includes clonal bone marrow plasma cells ($\geq 10\%$) or biopsy proven Plasmacytoma, with one or more 'Myeloma defining event'. An elevated involved : uninvolved serum FLC ratio >100 (and involved FLC $\geq 100\text{mg/L}$ is now considered a 'Myeloma defining event'.

Comment

Measurement of free light chain (FLCs) aids in the diagnosis & monitoring of Multiple myeloma, Waldenstrom's macroglobulinemia, AL Amyloidosis & Light chain deposition disease. Serum concentration of FLCs is dependent upon the balance between production by plasma cells & their progenitors and renal clearance.



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When there is increased polyclonal immunoglobulin production and/or renal impairment, concentration of both kappa & lambda light chains can increase by 30-40 fold but the Kappa / Lambda ratio remains unchanged. In contrast, tumor produces monoclonal excess of only 1 type of light chains, often with bone marrow suppression of alternate light chains resulting in abnormal kappa/lambda ratio.			

The combination of Serum protein electrophoresis (SPE) & Immunofixation (IFE) are highly sensitive but fail to identify a few patients with Light Chain Multiple Myeloma & all patients of Non Secretory Multiple Myeloma. Combination of free light chain assay with SPE & IFE improves the sensitivity to 99%.

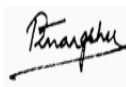
Uses

As per recommendations by IMWG:

- FLC,SPE and serum IFE for screening
- For diagnosis and prognosis of all patients with MGUS, Smoldering or active multiple myeloma, Solitary plasmacytoma and AL amyloidosis
- Serial measurements routinely in patients with AL amyloidosis and Multiple myeloma
- patients with oligosecretory disease
- In all patients who have achieved a complete response (CR) to assess for stringent CR


DMC - 67327

Dr Anjalika Goyal
MD,Biochemistry
Consultant Biochemist
NRL - Dr Lal PathLabs Ltd


DMC - 89819

Dr Himangshu Mazumdar
MD, Biochemistry
Sr. Consultant Biochemist
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DMC - 9550

Dr Nimmi Kansal
MD, Biochemistry
Technical Director - Clinical
Chemistry & Biochemical Genetics
NRL - Dr Lal PathLabs Ltd


MCI 15-19066

Dr.Richa Sirohi
MD, Biochemistry
Sr. Consultant Biochemist
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Test Report

Test Name	Results	Units	Bio. Ref. Interval
MULTIPLE MYELOMA MONITORING PANEL			
ALBUMIN, SERUM (BCG)	4.42	g/dL	3.20 - 4.60
CREATININE, SERUM (Modified Jaffe, IDMS traceable)			
Creatinine	1.82	mg/dL	0.70 - 1.30
GFR Estimated	40	mL/min/1.73m2	>59
GFR Category	G3b		

Advise

1. CKD Risk Map (Z1014)
2. Cystatin C, serum (B173)

Note

1. GFR, estimated (eGFR) calculated using the 2021 CKD-EPI creatinine equation and GFR Category reported as per KDIGO guideline 2012.
2. eGFR category G1 or G2 does not fulfil the criteria for CKD, in the absence of evidence of kidney damage

CALCIUM, SERUM (Arsenazo III)	9.60	mg/dL	8.80 - 10.20
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MULTIPLE MYELOMA MONITORING PANEL

LDH;LACTATE DEHYDROGENASE, SERUM (IFCC)	201.00	U/L	120 - 246
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Comments

Lactate dehydrogenase (LDH) is a nonspecific enzyme found in most organs. Highest concentrations are found in liver, heart, kidney and blood cells. LDH measurements are used in the diagnosis and treatments of liver diseases like Acute viral hepatitis, Cirrhosis & Metastatic carcinoma; Cardiac diseases like Myocardial infarction; Tumors of lungs / kidneys & Hematologic disorders like Megaloblastic anemia & Hemolytic anemia.



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
UREA, SERUM (Urease GLDH)			
Urea	38.00	mg/dL	17.00 - 49.00
BUN; Blood Urea Nitrogen	17.75	mg/dL	8.00 - 23.00

Page 9 of 13

If Test results are alarming or unexpected, client is advised to contact the Customer Care immediately for possible remedial action.
Tel: 011-4988-5050, Fax: +91-11-2788-2134, E-mail: customer.care@lalpathlabs.com

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Test Report

Test Name	Results	Units	Bio. Ref. Interval
COMPLETE BLOOD COUNT;CBC (Photometry, electrical impedance, vcs technology & calculated)			
Hemoglobin	11.90	g/dL	13.00 - 17.00
Packed Cell Volume (PCV)	35.40	%	40.00 - 50.00
RBC Count	3.90	mill/mm3	4.50 - 5.50
MCV	91.00	fL	83.00 - 101.00
Mentzer Index	23.3		
MCH	30.40	pg	27.00 - 32.00
MCHC	33.50	g/dL	31.50 - 34.50
Red Cell Distribution Width (RDW)	14.70	%	11.60 - 14.00
Total Leukocyte Count (TLC)	4.30	thou/mm3	4.00 - 10.00
Differential Leucocyte Count (DLC)			
Segmented Neutrophils	40.30	%	40.00 - 80.00
Lymphocytes	28.00	%	20.00 - 40.00
Monocytes	27.30	%	2.00 - 10.00
Eosinophils	4.00	%	1.00 - 6.00
Basophils	0.40	%	<2.00
Absolute Leucocyte Count			
Neutrophils	1.73	thou/mm3	2.00 - 7.00
Lymphocytes	1.20	thou/mm3	1.00 - 3.00
Monocytes	1.17	thou/mm3	0.20 - 1.00
Eosinophils	0.17	thou/mm3	0.02 - 0.50



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
Basophils	0.02	thou/mm3	0.02 - 0.10
Platelet Count	167	thou/mm3	150.00 - 410.00
Mean Platelet Volume	9.7	fL	6.5 - 12.0
There is monocytosis Followup and clinical correlation			

Comment

In anaemic conditions Mentzer index is used to differentiate Iron Deficiency Anaemia from Beta- Thalassemia trait. If Mentzer Index value is >13, there is probability of Iron Deficiency Anaemia. A value <13 indicates likelihood of Beta- Thalassemia trait and Hb HPLC is advised to rule out the Thalassemia trait.

Note

- As per the recommendation of International council for Standardization in Hematology, the differential leucocyte counts are additionally being reported as absolute numbers of each cell in per unit volume of blood
- Test conducted on EDTA whole blood



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Test Name	Results	Units	Bio. Ref. Interval
LIVER PANEL 1; LFT,SERUM (Spectrophotometry)			
AST (SGOT)	22.0	U/L	19.00 - 48.00
ALT (SGPT)	12.0	U/L	10.00 - 49.00
AST:ALT Ratio	1.83		<1.00
GGTP	12.0	U/L	0 - 73
Alkaline Phosphatase (ALP)	97.00	U/L	30.00 - 120.00
Bilirubin Total	0.41	mg/dL	0.20 - 1.10
Bilirubin Direct	0.10	mg/dL	<0.3
Bilirubin Indirect	0.31	mg/dL	<1.10
Total Protein	6.80	g/dL	5.70 - 8.20
Albumin	4.42	g/dL	3.20 - 4.60
Globulin(Calculated)	2.38	gm/dL	2.0 - 3.5
A : G Ratio	1.86		0.90 - 2.00

Note

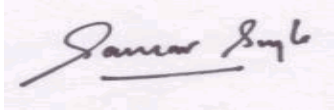
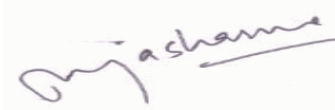
1. In an asymptomatic patient, Non alcoholic fatty liver disease (NAFLD) is the most common cause of increased AST, ALT levels. NAFLD is considered as hepatic manifestation of metabolic syndrome.
2. In most type of liver disease, ALT activity is higher than that of AST; exception may be seen in Alcoholic Hepatitis, Hepatic Cirrhosis, and Liver neoplasia. In a patient with Chronic liver disease, AST:ALT ratio>1 is highly suggestive of advanced liver fibrosis.
3. In known cases of Chronic Liver disease due to Viral Hepatitis B & C, Alcoholic liver disease or NAFLD, Enhanced liver fibrosis (ELF) test may be used to evaluate liver fibrosis.
4. In a patient with Chronic Liver disease, AFP and Des-gamma carboxyprothrombin (DCP)/PIVKA II can be used to assess risk for development of Hepatocellular Carcinoma.



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Test Report

Test Name	Results	Units	Bio. Ref. Interval
			
Dr. Gaurav Singla MBBS MD-Pathology Consultant Pathologist	Dr Pooja Sharma MD, Pathology Chief of Laboratory Dr Lal PathLabs Ltd		

-----End of report -----



IMPORTANT INSTRUCTIONS

•Test results released pertain to the specimen submitted. •All test results are dependent on the quality of the sample received by the Laboratory. •Laboratory investigations are only a tool to facilitate in arriving at a diagnosis and should be clinically correlated by the Referring Physician. •Report delivery may be delayed due to unforeseen circumstances. Inconvenience is regretted. •Certain tests may require further testing at additional cost for derivation of exact value. Kindly submit request within 72 hours post reporting. •Test results may show interlaboratory variations. •The Courts/Forum at Delhi shall have exclusive jurisdiction in all disputes/claims concerning the test(s) & or results of test(s). •Test results are not valid for medico legal purposes. •This is computer generated medical diagnostic report that has been validated by Authorized Medical Practitioner/Doctor. •The report does not need physical signature.
(#) Sample drawn from outside source.
If Test results are alarming or unexpected, client is advised to contact the Customer Care immediately for possible remedial action.
Tel: +91-11-49885050,Fax: - +91-11-2788-2134, E-mail: lalpathlabs@lalpathlabs.com





Postgraduate Institute of Medical Education and Research, Chandigarh

Department of Hematology

Flow cytometric assay for minimal residual disease (MRD)

Name V K Gurg Age/sex 67 Yr/M CR No. 201405136462
Ward/OPD Clinical Hematology & Medical Oncology-Hematology Clinic

Requisition Date	Lab ID	Sample	Consultant	BMA No.
18/02/2025	MRD-175/25	Bone Marrow Aspirate	-	A-473/25

TLC ($\times 10^9$ /L)	Neutrophils (%)	Lymphocytes (%)	Monocytes (%)	Eosinophils (%)	Other (%)
6.13	-	-	-	-	-

Bone Marrow Diagnosis

Multiple myeloma, on therapy

Gated events

SSC vs CD138 vs CD319 - 0.07% of all viable events.

RESULT

Percentage of MRD (%) Nil
Number of events acquired 4.8 million
Cellularity of the sample Adequate
Viability of the sample (%) 86%

Comments

No residual clonal plasma cells detected.

Note - 0.07% normal plasma cells are noted with a kappa:lambda ratio of 1.5:1.

Advice

-

Senior Resident Dr. Namrata Kaul Faculty Dr. Nabhajit Mallik
Typed by Ms. Navneet Kaur Validated on 20-02-25 09:43 AM
Technologist Mrs. Parveen Bose Validated by Nabhajit Mallik(Assistant Professor)



Department of Hematology
Postgraduate Institute of Medical Education & Research, Chandigarh
REPORT OF BONE MARROW ASPIRATION / TREPHINE BIOPSY

Name: V K Gurg **Age / Sex:** 67 Yr/M **CR No:** 201405136462 **B.M.No:** A-473/25
Cl.I /c: **Ward:** Clinical Hematology & Medical Oncology- Hematology Clinic **Dated:** 17/02/2025

Clinical Case of multiple myeloma on Daratumumab VRD. Now presented with relapse and is on
Diagnosis:- Daratumumab KPD. No hepatomegaly splenomegaly or lymphadenopathy. SPEP = 0.1, SIFE = IgG-kappa, SFLC = Kappa:lambda - 20.1.

HEMOGRAM DETAILS

Hemogram No: H-125

HB:- 10.4 **gm/dl** **Retic:** 0.93 (c) % **PLT:** 139 **X10⁹/L** **TLC:** 6.13 **X10⁹/L**
DLC P- 62 L- 20 M- 16 E- 2 Ba- - BI- - Pm-- My- - Mm-- nRBC- -

PBF: Mild anisopoikilocytosis. Normocytic normochromic red cells admixed with microcytes, tear drop cells, few elliptocytes and schistocytes (nearly 1 percent). Platelets are adequate with few large forms. No rouleaux formation or circulating plasma cells seen.

BONE MARROW FINDINGS

Particles: Particulate **NE:E Ratio** - **M:E Ratio** 2:1
Cellularity: Normocellular -

Blasts:	2	Erythropoiesis:	
Promyelocytes:	1		Normoblastic
Myelocytes:	10		-
Metamyelocytes:	5	Thrombopoiesis:	
Polymorphs:	24		Adequate
Lymphocytes:	26		normal morphology
Monocytes:	-	Others:	
Eosinophils:	-		Erythroid precursors -23%, Eo-Baso precursors - 4%.
Basophils:	-		
Plasma Cells:	5		

CYTOCHEMISTRY

LAP - **MPO** - **PAS** - **PERLS** 4+

TREPHINE BIOPSY REPORT

Trephine Biopsy No : Tx-373/25

Bilateral trephine biopsies measuring 2.2 and 2.3 cm show normocellular marrow spaces. There is adequate representation of the normal hematopoietic elements. No interstitial excess or aggregates of plasma cells are seen.

Reticulin - -

Interpretation Known case of multiple myeloma, on Daratumumab: Normocellular marrow shows 5% plasma cells.

Advice Kindly collect flow cytometry MRD report which will be available online.

JR Dr. Aditya **SR** Dr. Shruti Madan **Faculty** Dr. Jasmina Ahluwalia
Validated On 24-02-25 08:46 AM **Validated By** Jasmina Ahluwalia(Professor)